A Project Report

On

FitCore: Integrated Fitness and Workout Management Portal

In Partial Fulfilment of Bachelor of Computer Applications (B.C.A) Gujarat University

Developed By

Bhargavi Radadiya(6094)
Anjali Sutariya(6112)
Tanvi Vaghasiya(6120)
(Sem-V Bachelor of Computer Applications)

FitCore: Integrated Fitness and Workout Management Portal

Definition:

The **FitCore: Integrated Fitness and Workout Management Portal** is a web-based platform developed to simplify the management of residential fitness centers and personal training operations. It enables fitness administrators and coaches to manage members, workout programs, diet plans, coach schedules, payment transactions, and reward systems through a centralized digital system.

Overview:

- Fitness communities and commercial gym members often face challenges such as unstructured workout plans, manual calorie tracking, poor communication between members and fitness coaches, and inefficient slot scheduling.
- Traditional paper-based log sheets are time-consuming, prone to errors, and make it difficult to maintain accurate historical records and track physical progress.
- The FitCore: Integrated Fitness and Workout Management Portal is designed to digitize and automate workout and nutrition tracking activities through a centralized web-based platform.
- It enables members to log daily workouts, log nutritional food intake, view training slots, book appointments with coaches, track progress metrics, and access fitness-related resources from anywhere at any time.
- The system provides real-time updates on appointment status and physical metrics, allowing members to monitor the progress of their fitness journey without repeated manual logs.
- It also maintains a complete history of workouts, diets, and measurements, ensuring proper documentation and accountability.
- The portal enhances communication by enabling coaches to publish announcements, workout assignments, and nutrition plans that are instantly accessible to their clients.
- This ensures that fitness goals and instructions reach every member of the community quickly and effectively.
- Role-based authentication ensures that members, coaches, and administrators have access only to the features relevant to their responsibilities, thereby improving system security and protecting sensitive personal health information.

Objectives

The main objectives of the **FitCore: Integrated Fitness and Workout Management Portal** are:

- To provide a centralized platform for managing members, coaches, appointments, workouts, diets, and payments.
- To improve communication between fitness coaches and members through direct booking and message channels.
- To automate the day-to-day management of residential fitness centers and gyms.
- To provide a centralized platform for managing members, coaches, appointments, workouts, diets, and payments.
- To simplify workout program creation and track the physical progress of members.
- To improve communication between fitness coaches and members through direct booking and message channels.
- To maintain accurate records of member profiles, daily food logs, and exercise progress.
- To reduce paperwork and manual record keeping.
- To improve transparency, accountability, and efficiency in coaching and gym management.
- To provide secure access based on user roles such as Administrator, Coach, and Member.

Existing System:

Currently, many gyms and fitness centers maintain records manually using registers, paper cards, and spreadsheets. Workout logging, diet planning, appointment scheduling, and fee tracking are handled manually.

Disadvantages of Existing System:

- Paper-based record management is time-consuming.
- Workouts and diets are difficult to track and progress is hard to measure.
- High possibility of data loss or record misplacement (e.g., losing workout cards).
- Membership payment and renewal records are difficult to maintain.
- Communication between members and fitness coaches is slow.
- Coach slots and appointments are managed manually, causing double-bookings.
- No centralized database for fitness information.
- Generating fitness reports and client tracking requires significant manual effort.
- Increased chances of human errors in calculations (e.g. daily macros/calories).

Proposed System

The proposed system is a web-based portal that provides a centralized platform for managing all fitness-related operations digitally. It automates workout logging, diet tracking, coach-client scheduling, progress analysis, and membership management.

The system offers secure login facilities for administrators, coaches, and members and stores all information in a centralized database.

Features

- Secure Login System
- Member Profile & Health Management
- Workout Program Logging & Tracking
- Diet Plan & Macronutrient Tracking
- Coach Allocation & Slot Booking
- Stripe Payment Integration
- Dashboard with Reports & Dynamic Charts
- Role-Based Access Control (Admin, Coach, Member)
- Search and Filter Facilities

1. Admin Module

- Admin Login
- Dashboard with Analytical Reports
- Manage Coaches & Members
- Manage Membership Tiers
- Manage Exercise Database
- Manage Payment Records
- Generate Activity & Payment Reports (CSV Export)

2. Member Module

- Member Registration
- Member Login
- View Personal Dashboard (Daily targets, progress charts)
- Log Daily Workout Session (Sets, Reps, Weight)
- Log Food Intake & Track Macronutrients (Calories, Protein, Carbs, Fats)
- Book Appointments with Coach (General coach)
- Hire personal Coach (Personal coach)
- Make Payment
- Update Physical Metrics (Weight, Waist, Arms, Photos)

3. Workout Module

- Create Exercise Templates
- Build Customized Routine Workout Programs (by Coaches)
- Track Daily Exercise Completion
- Analyze Progress with Metrics (Weight lifted, sets done)
- View Workout History Logs

4. Appointment Module

- Set Coach Availability Hours (Day of week slots)
- Request Appointment Booking
- Approve or Decline Bookings
- Update Session Status (Pending, Confirmed, Completed, Cancelled)
- Manage Availability Slots dynamically

5. Diet Module

- Create Diet Plans with Macronutrient Goals
- Log Meal Intake (Breakfast, Lunch, Dinner, Snacks)
- Calculate Daily Calorie Consumption
- Monitor Protein, Carb, and Fat Progress Rings
- Generate Nutrition Consumption Reports

6. Coach Module

- Coach Registration
- Coach Login
- View Coach Dashboard (Summary of clients, pending appointments)
- Manage Assigned Clients
- Create & Assign Workout Programs
- Create & Assign Diet Plans
- Manage Schedule Availability (Time Slots)
- View & Update Appointment Status
- View Client Progress Logs (Workouts, Diet, Photos)

7. User Management Module

- User Registration (Member / Coach / Admin)
- Login Authentication & JWT Validation
- Change Password / Reset Password Flow
- Update Personal Profile & Health Details
- Role Management & Authorization Guards

Technology Stack

Component	Technology
Frontend	Next.js, Tailwind CSS, TypeScript
Backend	Node.js, Express.js
Database	MongoDB
Server	Node.js
IDE/Editor	Visual Studio Code
Database Tool	MongoDB Compass / MongoDB Atlas
Browser	Google Chrome
Operating System	Windows 10/11
Version Control	Git & GitHub